

Two Displays, One Data Set

AP Statistics · Unit 1 · Topic 1.8 · B: 2026-09-22 · E: 2026-09-28 · TEACHER KEY

CED TETHER

- Enduring Understanding: UNC-1 Graphical representations and statistics allow us to identify and represent key features of data.
- Skills:
- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- Learning Objectives and Essential Knowledge:
- UNC-1.L Represent summary statistics for quantitative data graphically. [Skill 2.B]
- **UNC-1.L.1** — Taken together, the minimum data value, the first quartile (Q1), the median, the third quartile (Q3), and the maximum data value make up the five-number summary.
- **UNC-1.L.2** — A boxplot is a graphical representation of the five-number summary (minimum, first quartile, median, third quartile, maximum). The box represents the middle 50% of data, with a line at the median and the ends of the box corresponding to the quartiles. Lines ("whiskers") extend from the quartiles to the most extreme point that is not an outlier, and outliers are indicated by their own symbol beyond this.
- UNC-1.M Describe summary statistics of quantitative data represented graphically. [Skill 2.A]
- **UNC-1.M.1** — Summary statistics of quantitative data, or of sets of quantitative data, can be used to justify claims about the data in context.
- **UNC-1.M.2** — If a distribution is relatively symmetric, then the mean and median are relatively close to one another. If a distribution is skewed right, then the mean is usually to the right of the median. If the distribution is skewed left, then the mean is usually to the left of the median.

Essential question: What information survives when individual observations are compressed into a boxplot?

DOK-3 skill: 4.B · **FRQ pattern:** what-a-boxplot-can-and-cannot-show-adjudicate

DOK rationale: Part (c) requires adjudicating two interpretations by coordinating the boxplot with the dotplot, distinguishing summaries from distribution features, and defending which display answers a decision question; no single calculation determines the response.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Read Omar's and Nia's claims aloud without confirming either.
- Video follow-along from minute 5 (five-number summaries and boxplots).
- Board slide back up at minute 33. Circulate; ask students to point before they name a feature.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to Omar or Nia from the two displays before the video.
- Complete the video follow-along on five-number summaries and boxplots.
- Finish (a)–(c); complete the exit line; turn in.

Questions to ask

- Point to 32 minutes on the dotplot. How many dots are nearby?
- Could a different data set have this same five-number summary but no gap?
- Which features disappeared when the 24 dots became five numbers?
- For the coach's decision, do you need a center or the number of students above 30?

Adult role

Solo teacher. Circulate 5 pairs. Do not say 'bimodal' or confirm Nia; ask students to locate the median on both displays and name what disappears.

[WARN] WATCH FOR

- Treating the median as a mode or as a value an observation must equal.
- Claiming the boxplot displays the two clusters because its box is wide.
- Calling 55 an outlier without checking the 1.5 IQR fence.

SCORING PART (c) — E / P / I

Expected elements

- **chooses-nia-with-gap** (required) — Chooses Nia and cites that the median of 32 lies in the dotplot's 27–37 minute gap, with no observations nearby
- **distinguishes-display-capabilities** (required) — Explains that the boxplot preserves center and spread but not the clusters, gap, or modality
- **chooses-dotplot-for-decision** (required) — Chooses the dotplot for the more-than-30-minute decision and uses the two groups or 12-of-24 split to justify it
- **names-boxplot-mislead** (required) — Explains that the boxplot alone can make the data look like one continuous group centered near 32

E	Chooses Nia, cites the empty interval around the median, accurately contrasts what the displays retain, and chooses the dotplot for the coach with the 12-of-24 split and the boxplot's misleading continuous-pile impression explained.
P	Chooses Nia and cites the gap but gives only a general display comparison, OR chooses the dotplot with no count or consequence, OR provides the capabilities and consequence but never commits to an interpretation.
I	Chooses Omar because 32 is the median, claims a boxplot shows the two clusters, or chooses a display with no feature from the data.

Common mistakes

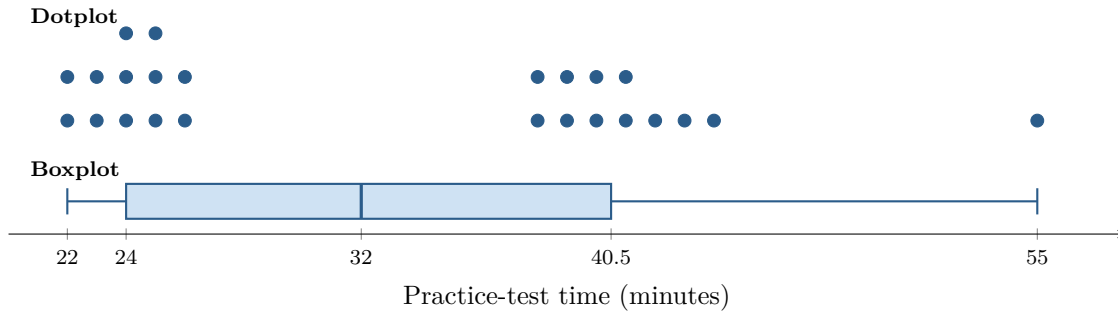
- Treating the median as the most common value; no student took 32 minutes
- Calling the empty 27–37 interval a short bar even though the display is a dotplot
- Saying a boxplot shows every observation because it shows the minimum and maximum
- Calling 55 a formal 1.5 IQR outlier; the upper fence is 65.25
- Choosing the boxplot for the coach merely because it is more compact

[STAR] TODAY'S PROBLEM — annotated

The displays show practice-test times, in minutes, for 24 students. Two students interpret the boxplot.

Omar: “Most students took about 32 minutes—the middle of the box.”

Nia: “A boxplot cannot tell us whether most students took about 32 minutes.”



FIRST TAKE (5 min)

Who do you trust more, Omar or Nia? Point to one feature in either display.

Key: Not graded. Expect Omar; the empty interval around 32 is the reveal.

Rules callout “DOTPLOTS AND BOXPLOTS (keep on the page)” is printed on the student sheet.

DOK 1 (a) Read the median and the first and third quartiles from the boxplot. What percent of the times lie in the box from Q1 to Q3?

Key: Median = 32 min, Q1 = 24 min, and Q3 = 40.5 min. The box contains the middle 50% of the times. The full five-number summary is (22, 24, 32, 40.5, 55).

DOK 2 (b) Describe the dotplot in context: include the two clusters, the gap, and the isolated high value. State two features that the boxplot preserves.

Key: Two clusters: 12 times from 22–26 min and 11 from 38–44 min; a 27–37 min gap; isolated 55 min. Ordered data: 22, 22, 23, 23, 24, 24, 24, 25, 25, 25, 26, 26, 38, 38, 39, 39, 40, 40, 41, 41, 42, 43, 44, 55. The boxplot preserves median 32, quartiles 24 and 40.5, and spread 22–55; it hides the clusters and gap.

DOK 3 (c) Whose interpretation is better supported? Cite the dotplot feature that settles it. Explain what a boxplot can and cannot show here. A coach asks whether most students need more than 30 minutes: choose a display, and explain what the boxplot alone would lead the coach to believe.

Key: Nia. No dot is near 32; the median is halfway between 26 and 38 in the gap, so 32 is not a typical cluster time. A boxplot shows center, quartiles, spread, and flagged outliers but hides gaps, clusters, and modality. Use the dotplot: 12 times are above 30 and 12 below, in two groups. The boxplot alone suggests one continuous group around 32.

EXIT

One thing the video changed about my first take: _____